

YUQUAN (Olivia) LIU

Current undergraduates student at the University of Maryland pursuing a degree in Computer Science

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EXPERIENCE

UMD FIRE Program - Climate Computing Stream

August 2025 - Current

Undergraduate Researcher

- Conducted critical analysis of scholarly articles to extract key arguments, methodologies, and findings
- Evaluated credibility & relevance and synthesized information from multiple sources using structured research criteria to support research discussions
- Collaborated with team members to analyze environmental datasets and provide constructive feedback during consistent discussions.
- Currently working with executing data processing tasks on high performance computing (HPC) infrastructure supporting climate simulations, investigating in large-scale model analysis.
- Experience with Linux/UNIX environments using command-line interface, while leveraging climate model simulation through HPC systems.

MIT Beaver Works Summer Institute

2024 Spring

Autonomous RACECAR Grand Prix Track - Role

- Designed modular program components using object-oriented principles (OOP)
- Collaborated in teams using shared codebases to implement and refine vehicle behaviors
- Tested and debugged algorithms in a 3D simulation environment
- Engineered Python-based control logic, strengthened foundations in applied robotics programming and system integration

MITHacker Blueprint 2024

November 2024

Full-Stack Developer

- Implemented full-stack features using Python and Javascript
- Structured modular code for both frontend and backend functionality
- Build and integrated pixel assets using Tiled into application interface
- Coordinated with other members on design, implementations, and debugging workflows.
- Demonstrated completed prototype through live technical presentation.

Past Projects

- Various Java data structure implementation. (Linked list, Trees, Graphs, etc)
- Projects utilizing file I/O, exception handling, and user interaction
- Various algorithm analysis
- Robotics visual perception implementation through python
- Javascript backend implementation

AWARDS

WHS Science National Honors Society

WHS Mu Alpha Theta Math Honors Society

AP Scholar with Distinctions

SKILLS

Programming Languages

Python, Java, Javascript, HTML, CSS

Adobe Applications

Adobe Audition, PhotoShop

EDUCATION

University of Maryland, College Park

Bachelor of Science in Computer Science

August 2025 - Current

Cumulative GPA: 3.83

Windham High School, Windham, New Hampshire

Graduated as Top 20 Students with Honorable Awards

August 2021 - May 2025

Cumulative GPA: 4.0

LANGUAGES

Mandarin - Native Fluency

English - Native Fluency